Metro Flux

1. Station Information — "Where are the stations, and what's there?"

StopPoint API

Link: api.tfl.gov.uk/StopPoint

This is your master list of stations. Send it a station ID and it hands back the name, GPS coordinates, which lines stop there, and accessibility info (lifts, step-free access, etc.). This is the foundation of your whole system — every station "node" in your app comes from here.

TfL Station Locations (map file)

Link: data.london.gov.uk/dataset/tfl-station-locations-2zmpy

A simple downloadable file (KML format) with every station's location already plotted. Handy if you want to quickly drop station pins onto a map without writing code to call the API first.

TfL Station Facilities

Link: data.london.gov.uk/dataset/tfl-station-facilities

Details on what's actually inside each station — toilets, ticket halls, lifts, escalators. Useful if your guidance system ever needs to route someone with accessibility needs away from a crowded platform.

Step-Free Access Guide

Link: step-free-access-and-toilet-data-guide.pdf

Not data itself — this is the instruction manual explaining how TfL structures its accessibility data, so you know how to read it correctly.

NaPTAN (station ID system)

Link: naptan.app.dft.gov.uk

This isn't a TfL resource — it's a UK government database of ID codes used for every public transport stop in the country. Good to know it exists because TfL's own APIs (especially the Crowding API below) use these IDs internally.

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2. Metro Network — "How do the lines connect?"

Line API

Link: api.tfl.gov.uk/Line

Gives you the tube lines themselves and which stations sit on each one, in order. This is what you'll use to build the "map" your load-balancing logic runs on — knowing that Oxford Circus connects the Central, Victoria, and Bakerloo lines, for example.

TfL GIS Open Data Hub

Link: gis-tfl.opendata.arcgis.com

A free library of downloadable map files (lines, stations, zones) in formats like GeoJSON — ready to drop straight into a mapping tool for your app's visuals.

Official Tube Maps

Links: Standard Tube map (PDF) · Rail & Tube services map (PDF)

The classic tube map. Good for double-checking your app's network data actually matches reality, and useful for design mockups. (Note: fine for personal/project use — if you ever wanted to sell something using the map image itself, that needs a separate commercial licence.)

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3. Timetables — "When are trains supposed to run?"

TfL Timetable Listings

Link: data.london.gov.uk/dataset/tfl-timetable-listings

The official scheduled timetable, refreshed every week. Useful as a baseline — if trains are running way later than this schedule says they should, that's often when crowding starts building up.

Line Timetable Endpoint

Link: api.tfl.gov.uk/Line/{id}/Timetable/{fromStopPointId}

Same idea as above but queryable directly through the API — ask "what's the scheduled timetable for the Central line starting from Bank?" and get a direct JSON answer instead of downloading a whole file.

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4. Live APIs — "What's happening right now?" (the most important section)

Crowding API — your #1 resource

Link: `api.tfl.gov.uk/crowding/{StationID}/Live`

This is the star of the whole project. TfL calculates how busy every station is, updated every 5 minutes, based on anonymised Wi-Fi/gate data. It gives you a simple number from 0 to 1 (roughly: under 0.4 = quiet, 0.4–0.7 = busy, over 0.7 = very busy). There's

also a historical version (`/{DayOfWeek}`) showing typical crowding patterns for any day of the week — great for training a prediction model. Build your load-balancing logic around this.

Arrivals API ("Countdown")

Link: `api.tfl.gov.uk/StopPoint/{id}/Arrivals`

Tells you exactly when the next trains are due at a station, platform by platform. Since TfL doesn't give you live GPS positions of trains (see the "what you can't get" section), this is the closest thing to real-time train tracking you'll get.

Line Status API

Link: `api.tfl.gov.uk/Line/Mode/tube/Status`

Tells you if a line has "Good Service," delays, or is partly closed. Delays are one of the biggest causes of sudden crowding (everyone piles onto a different line), so feed this into your system.

Disruptions & Planned Works API

Link: `api.tfl.gov.uk/Line/Mode/tube/Disruption`

Details on what's actually wrong and which stations are affected — plus advance notice of planned engineering closures (weekend works, etc.) so you can prepare for predictable crowding, not just react to surprises.

Journey Planner API

Link: `api.tfl.gov.uk/Journey/JourneyResults/{from}/to/{to}`

Give it a start and end station, and it works out possible routes for you. You don't need to build your own pathfinding — just ask this API for route options, then re-rank them using your own crowding data.

Vehicle API — a quick heads up

Link: `api.tfl.gov.uk/Vehicle`

Worth knowing this exists, but it only works for buses, not tube trains. Don't expect to get live GPS positions of tube trains from here (or anywhere else — TfL doesn't publish that).

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5. Mapping Assets — "What can I use to build the visuals?"

What	Link	Format
Full map library	gis-tfl.opendata.arcgis.com	GeoJSON, KML, Shapefile
Station points layer	TfL stations map	GeoJSON
Station locations (simple)	tfl-station-locations-2zmpy	KML
Standard Tube map	content.tfl.gov.uk/standard-tube-map.pdf	PDF

Use these to skip the work of drawing your own map — import them straight into your front end.

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6. Data Standards (GTFS) — a quick note

You may have heard of **GTFS**, the worldwide standard format for transit data. Good to know: **TfL does not officially publish a GTFS feed** for the Underground. Everything comes in TfL's own JSON format through the Unified API instead (the endpoints listed above). If you want to use GTFS-based tools or libraries, you'd have to build your own converter on top of TfL's API — it's a decision worth making early since it affects what tools you can use.

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7. What You Won't Be Able to Get

Important so you don't design a feature around data that simply isn't public:

- **CCTV of stations or platforms** — not available. (TfL does have a public camera API, but it's only for road traffic cameras, not stations.)
- **Real-time passenger counts** — only historical, not live. You can get past data from the Rolling Origin & Destination Survey and yearly entry/exit figures, but nothing live.
- **How full an individual train carriage is** — not published anywhere.
- **Exact crowd density on a specific platform** — you only get a whole-station busyness score (the Crowding API above), not per-platform detail.
- **Internal TfL operations** — signalling, staff schedules, control-room decisions — none of this is public.

Bottom line: build your project around the **Crowding API** as your main real-time signal, and use Arrivals + Line Status + Disruptions as supporting context. Don't plan around CCTV, live train GPS, or per-carriage occupancy — TfL simply doesn't offer these publicly.

Quick-Start Checklist

1. Sign up for a free key at api-portal.tfl.gov.uk
2. Pull your station list from the **StopPoint API**
3. Pull your line/network structure from the **Line API**

4. Set up polling for the **Crowding API** (your core real-time signal) and **Arrivals API**
5. Add **Line Status** and **Disruptions** to catch delay-driven crowding
6. Use the **Journey Planner API** for alternative-route suggestions
7. Grab map layers from the **GIS Open Data Hub** for your UI
8. Remember: no CCTV, no live train GPS, no per-carriage data — the Crowding API is your best substitute for all three